

- 4 (a) (i) 1 cycle
 $T = 1.90 \text{ ms}$ C1
 $f = \frac{1}{T} = \frac{1}{(1.90 \times 10^{-3})} = 526.3 = 526 \text{ Hz}$ A1
 (Students are allowed to use one or more cycles.)

- (ii) $\phi = \frac{\Delta t}{T} \times 2\pi$ C1
 $\frac{4}{5}\pi = \frac{\Delta t}{T} \times 2\pi$
 $\Delta t = \left(\frac{4\pi}{5}\right)\left(\frac{T}{2\pi}\right) = \frac{2}{5}(1.90) = 0.76 \text{ ms}$
 $1.50 + 0.76 = 2.26 \text{ ms}$ A1

- (iii) $I = \frac{P}{4\pi r^2} \Rightarrow I \propto \frac{P}{r^2}$
 $\frac{I_1}{I} = \frac{P_1}{r_1^2} \times \frac{r^2}{P}$
 for $I_1 = I$
 $r_1^2 = \frac{P_1}{P} r^2$ C1
 $r_1 = \sqrt{\frac{0.25P}{P}} r = \sqrt{0.25} (120) = 60 \text{ cm}$ A1

- (b) (i) All the particles in a progressive wave oscillate with the same amplitude. B1
 The particles in a stationary wave oscillate with amplitudes that range from zero at the nodes to a maximum at the antinodes.
- (ii) All the particles within a wavelength of a progressive wave have different phases. B1
 All the particles between two adjacent nodes of a stationary wave have the same phase. Particles in adjacent segments have a phase difference of π radians.

